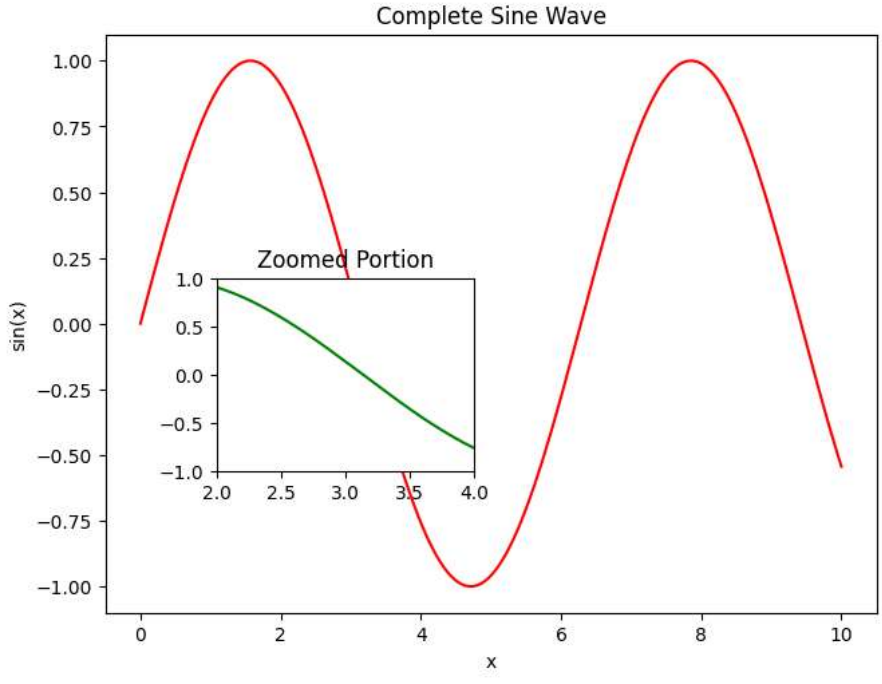
Inserting an axes object inside another axes object Write a Python program using Matplotlib to demonstrate how an Axes object can be inserted inside another Axes object. Create a main Axes to plot the function y=sin(x), 0≤ x≤10 and create a smaller inset Axes inside the main Axes using the add_axes() method. show only the portion from x=2 to x=4. Label the main plot appropriately and give suitable titles to both the main and inset plots.

```
import matplotlib.pyplot as plt
import numpy as np
x=np.linspace(0,10,200)
y=np.sin(x)
fig=plt.figure()
#main axes
ax1=fig.add_axes([0.1, 0.1, 0.9, 0.9])
ax1.plot(x,y,color='r')
ax1.set_xlabel('x')
ax1.set_ylabel('sin(x)')
ax1.set_title('Complete Sine Wave')
# Inset Axes
ax2 = fig.add_axes([0.23, 0.32, 0.3, 0.3])
ax2.plot(x, y, 'g-')
ax2.set_title('Zoomed Portion')
# Select only a small portion
ax2.axis([2, 4, -1, 1])
plt.show()
```



Write a Python program using NumPy and Matplotlib to generate a noisy sinusoidal signal and display it using a main Axes object. Create a second, smaller inset Axes object inside the main plot and use it to show an enlarged view of a selected portion of the noisy sinusoidal signal. The program should:

1. Generate a sinusoidal signal with random noise.
2. Create a Figure and a main Axes object using add_axes().
3. Plot the complete noisy signal in the main Axes.
4. Create a smaller Axes object inside the main Axes.
5. Plot the same signal in the inset Axes.
6. Set suitable axis limits in the inset Axes to display an enlarged portion of the original signal.
7. Add suitable labels and titles.

```
import numpy as np
import matplotlib.pyplot as plt

# Create data
x = np.linspace(0, 10, 500)
y = np.sin(x) + 0.2 * np.random.randn(500)

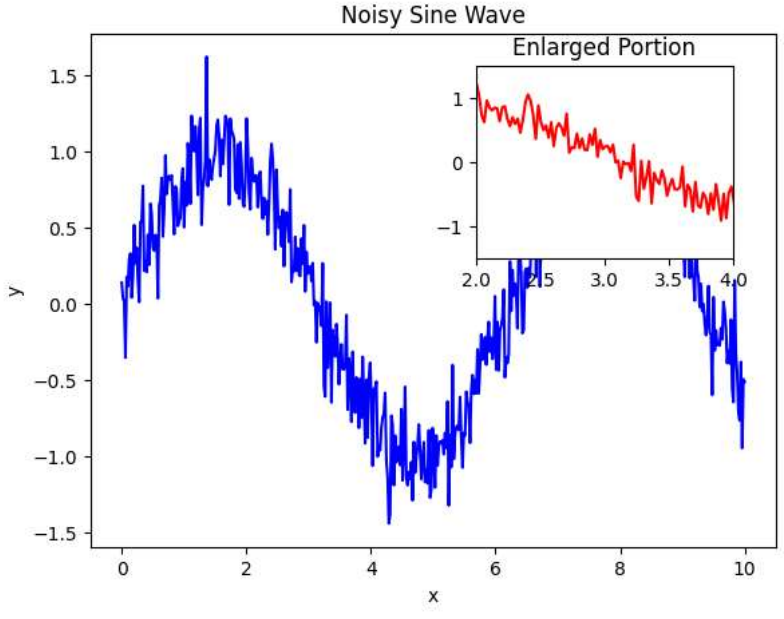
# Create Figure
fig = plt.figure()

# Main Axes
ax1 = fig.add_axes([0.1, 0.1, 0.8, 0.8])
ax1.plot(x, y, 'b-')
ax1.set_xlabel("x")
ax1.set_ylabel("y")
ax1.set_title("Noisy Sine Wave")

# Small Axes inside the main Axes
ax2 = fig.add_axes([0.55, 0.55, 0.3, 0.3])
ax2.plot(x, y, 'r-')
ax2.set_title("Enlarged Portion")

# Select the portion to enlarge
ax2.axis([2, 4, -1.5, 1.5])

plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(1, 10000, 100)
y = np.sqrt(x)

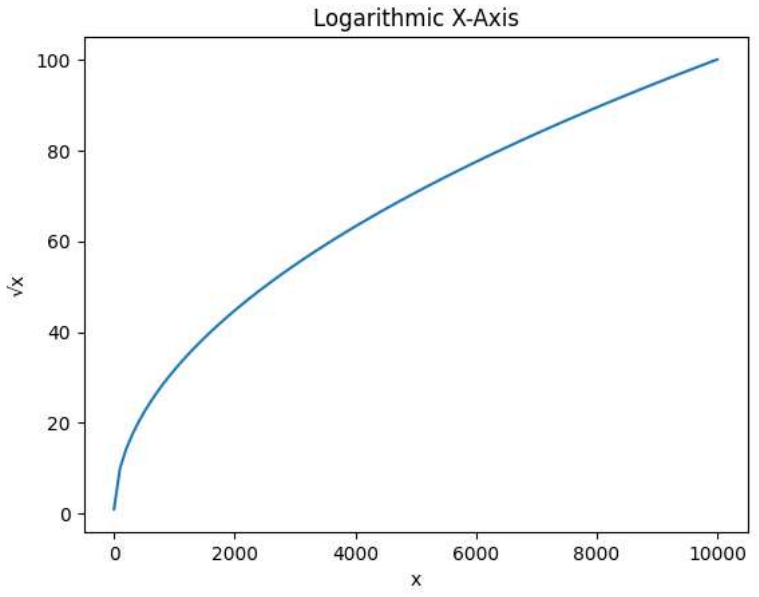
fig, ax = plt.subplots()

ax.plot(x, y)

#ax.set_xscale('log')

ax.set_xlabel('x')
ax.set_ylabel('√x')
ax.set_title('Logarithmic X-Axis')

plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(1, 10000, 100)
y = np.sqrt(x)

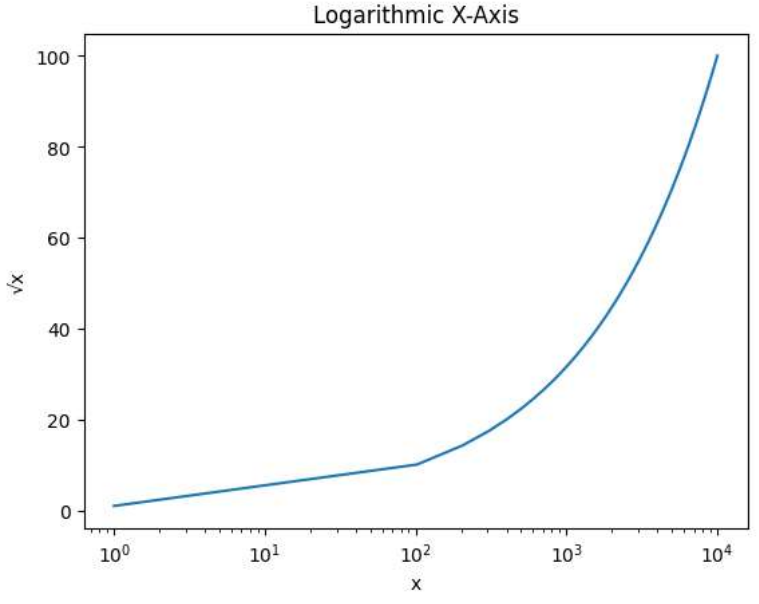
fig, ax = plt.subplots()

ax.plot(x, y)

ax.set_xscale('log')

ax.set_xlabel('x')
ax.set_ylabel('√x')
ax.set_title('Logarithmic X-Axis')

plt.show()
```



Write a Python program using NumPy and Matplotlib to plot the functions y1=(x−3)2 and y2=−(x−2)2+8 for (0≤x≤5). Use the fill_between() method to colour the region between the two curves where (y2≥ y1). Label both axes and

give the graph an appropriate title.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 5, 100)

y1 = (x - 3)**2
y2 = -(x - 2)**2 + 8

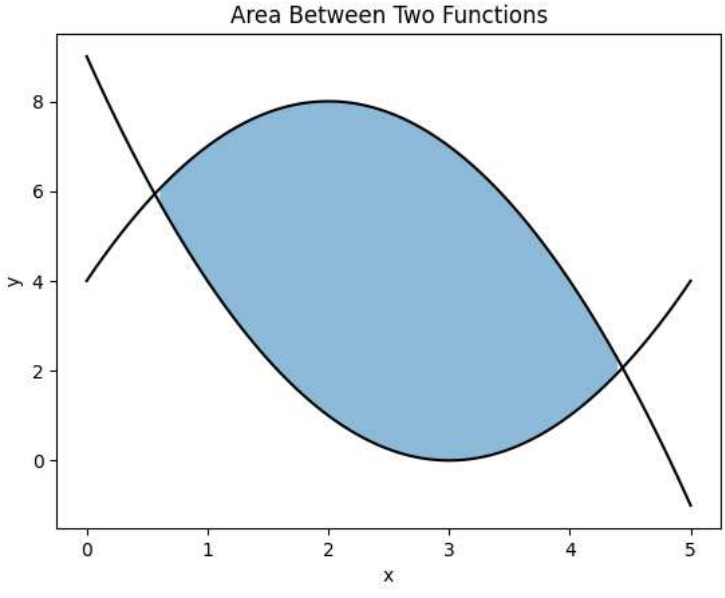
fig, ax = plt.subplots()

# Plot the two functions
ax.plot(x, y1, color='black')
ax.plot(x, y2, color='black')

# Colour the area between the curves
ax.fill_between(x, y1, y2, where=y2 >= y1,
                alpha=0.5) # the more the alpha the darker the colour

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Area Between Two Functions')

plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 5, 100)

y1 = x
y2 = 2*x

fig, ax = plt.subplots()

# Plot the two lines
ax.plot(x, y1)
ax.plot(x, y2)

# Fill the area between them
ax.fill_between(x, y1, y2, color='blue', alpha=0.3)

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Area Between Two Lines')
```

```
Text(0.5, 1.0, 'Area Between Two Lines')
```

